Lecture 5

Capacitor

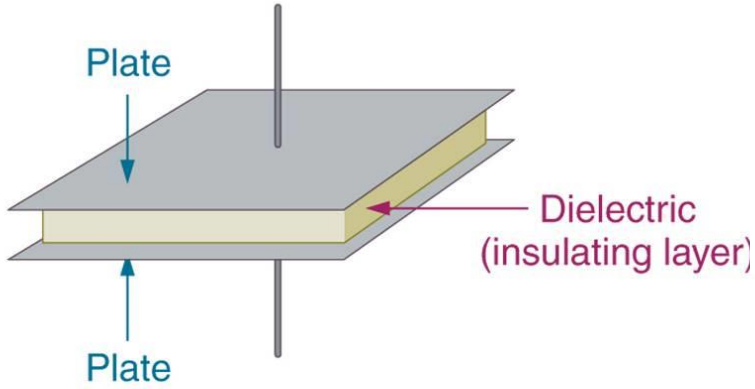
Slide Courtesy:
1.Electronics Technology Fundamentals, 3rd ed. Paynter and Boydell
2. Electronics Fundamentals 9th edition, Floyd/Buchla/Snyder

What is a Capacitor?

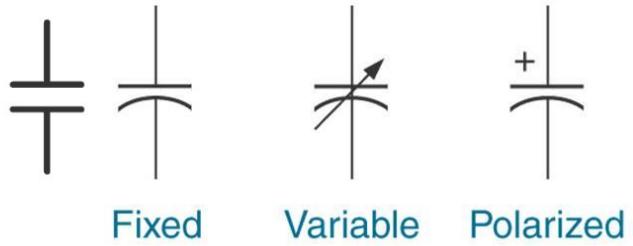
- A **capacitor** is a passive electronic component which has the ability or "capacity" to store energy in the form of an electrical charge, producing a potential difference (Static Voltage) across its plates.

Basic Structure

- It consists of two or more pieces of conducting material separated by an insulating material.
- The insulating layer between a capacitors plates is commonly called the **Dielectric**



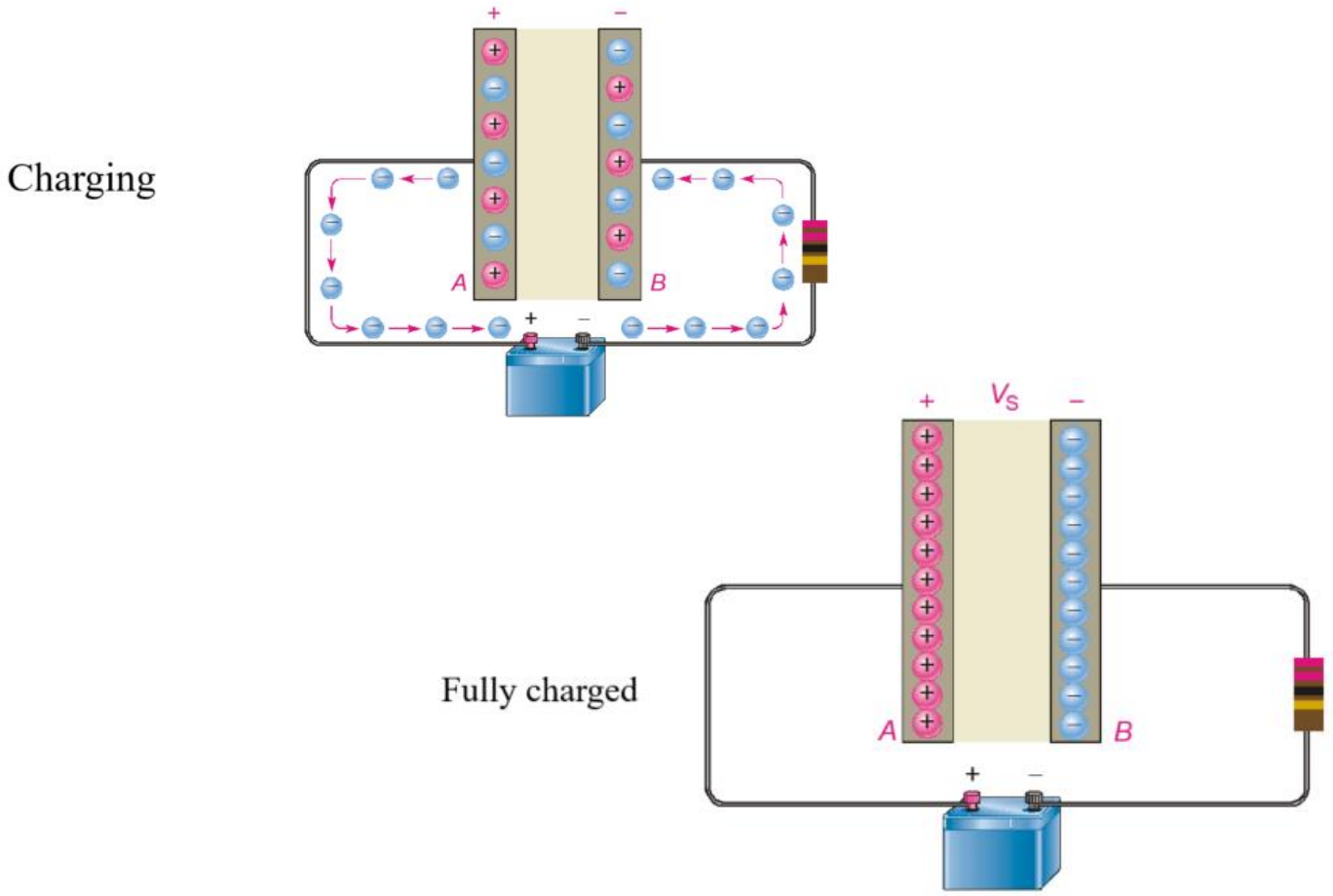
(a) Capacitor construction



(b) Common capacitor symbols

How Does a Capacitor Work?

- **Charging Process:**
  DC voltage applied across capacitor causes charges to accumulate on plates.
- Positive (+ve) charge gathers on one plate, negative (-ve) charge on the opposite plate.
- Charging continues until voltage across plates equals the applied voltage.
- At this point, the capacitor is **fully charged** with electrons.



A capacitor with stored charge can act as a temporary source of battery.

Capacitance

Capacitance – is the measure of the capacitors' ability to store charge.
The more charge per unit of voltage that a capacitor can store, the greater its capacitance.

$$C = \frac{Q}{V}$$

$$Q = CV$$

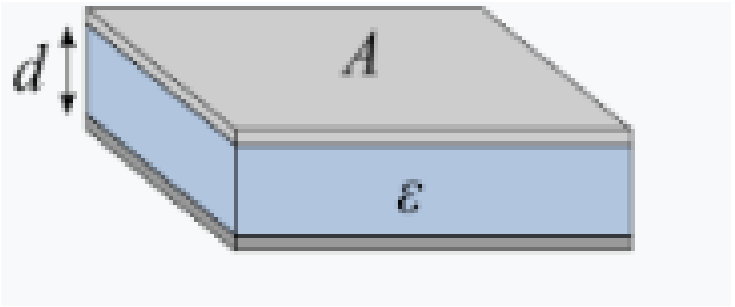
- Capacity – amount of charge that a capacitor can store per unit volt applied
- Capacity is directly proportional to charge and inversely proportional to voltage
- Rearranging, the amount of charge on a capacitor is determined by the size of the capacitor (C) and the voltage (V).

Unit of Measurement: Measured in **farads (F)**.
Most practical capacitors are in microfarads (μF), nanofarads (nF) or picofarads (pF).

Capacitance

Physical Factors Affecting Capacitance:

1. Area of Plates: Larger plates = higher capacitance.
2. Distance Between Plates: Closer plates = higher capacitance.
3. Dielectric Material: A better dielectric increases capacitance.
Dielectric constant (ε_r): It is the measure of the material's ability to establish an electric field.
e.g.: Vacuum: 1, Mica: 5 to 7, Quartz: 3.8, Paper: 2.5 – 3.5, oil: 4; Glass: 4.7 to 10



$$C = \varepsilon\left(\frac{A}{d}\right) \quad or \quad C = \varepsilon_0\left(\frac{\varepsilon_r A}{d}\right)$$

$$C = 8.85 \times 10^{-12}\left(\frac{\varepsilon_r A}{d}\right)$$

ε = Permittivity of the material

$\varepsilon = \varepsilon_0 * \varepsilon_r$

$\varepsilon_0 = 8.85 \times 10^{-12}$F/m (the permittivity of a vacuum)

$\boldsymbol{\varepsilon_r}$ = relative permittivity (dielectric constant) of the material;

Types of Capacitors

- **Non- polarized Capacitors:**
  - Orientation Doesn't Matter: These capacitors can be connected in either direction.
  - Lower (picofarads to microfarads)
  - e.g: Ceramic capacitors
- **Polarized Capacitors**
  - Must be connected with correct polarity
  - High (usually in microfarads to farads)
  - e.g: Electrolytic Capacitors

- **Ceramic Capacitors:**
  - High dielectric constants (typically 1200),
  - Typical capacitance value (1 pF to 2.2uF)
  - Non-polarized, small in size
  - Used in high-frequency applications
- **Electrolytic Capacitors:**
  - Larger capacitance (1 μF and up)
  - Polarized, used for power supply filtering
- **Tantalum Capacitors:**
  - High stability and reliability
  - Used in precision circuits.
- **Supercapacitors:**
  - Very high capacitance
  - Used for energy storage in backup power systems.



Capacitor Labeling

- **Capacitance Value:** Capacitors use several labeling methods.
- Large (physically) capacitors usually has its value printed directly on the case
- Smaller components are generally labeled using a two-digit or three-digit code
  - Two-digit code: the number represents the value of the component in pF. Example: **15** means 15 pF
  - Three-digit code: the code is interpreted like the first three digits of a resistor code.
  - **473** = 47 x 10^3 pF = 47 nF ; **104** = 100,000 pF = 0.1 µF

    Note: The numbers 6 and 7 are not used as multiplier values
    The numbers 8 and 9 are decoded as follows: 8 = 0.01 and 9 = 0.1; Example: 158 = 0.15 pF
- **Voltage Rating**: 50V indicates a capacitor can withstand 50 volts.
- **Tolerance:** Indicates variation from the labeled capacitance values. J: ±5% ;K: ±10%; M: ±20%







Series and Parallel Capacitors

Connecting Capacitors in Series:

- When capacitors are connected in series, the total capacitance is smaller than the smallest one.

Connecting Capacitors in Parallel:

- When capacitors are connected in parallel, the total capacitance is the sum of the individual capacitors.

Proof: Apply Kirchhoff's law and the capacitance equation (Q = CV)

- For series: $Q_T = Q_1 = Q_2 = ..$ And $Vs = V_1 + V_2 + ..$
- For parallel: $Q_T = Q_1 + Q_2 + ..$ And $Vs = V_1 = V_2 = ..$

Series: $C_T = \frac{1}{\frac{1}{C_1}+\frac{1}{C_2}+\cdots+\frac{1}{C_n}}$

Parallel: $C_T = C_1 + C_2 + \cdots + C_n$

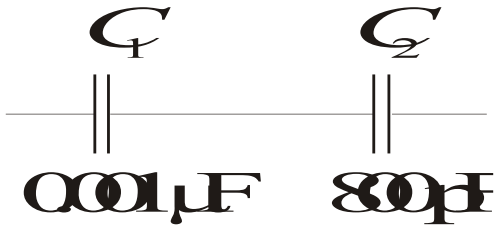
where,
C_T = the total series capacitance
C_n = the highest-numbered capacitor

Two Series Capacitors

The total capacitance of two series capacitors is: $C_T = \dfrac{1}{\dfrac{1}{C_1} + \dfrac{1}{C_2}}$

This reduces to the product-over-sum rule: $C_T = \dfrac{C_1 C_2}{C_1 + C_2}$

If a 0.001 μF capacitor is connected in series with an 800 pF capacitor, the total capacitance is 444 pF



If another 0.001 μF capacitor is connected in series with the capacitors, the total capacitance is 308 pF

Parallel Capacitors

When capacitors are connected in parallel, the total capacitance is the sum of the individual capacitors. The general equation for capacitors in parallel is

$$C_T = C_1 + C_2 + C_3 + \ldots + C_n$$

Example

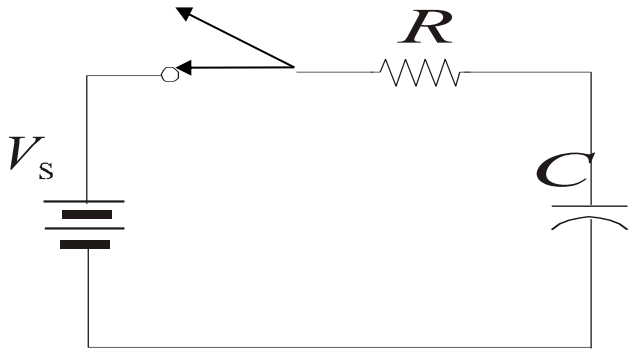
If a 0.001 μF capacitor is connected in parallel with an 800 pF capacitor, the total capacitance is 1800 pF

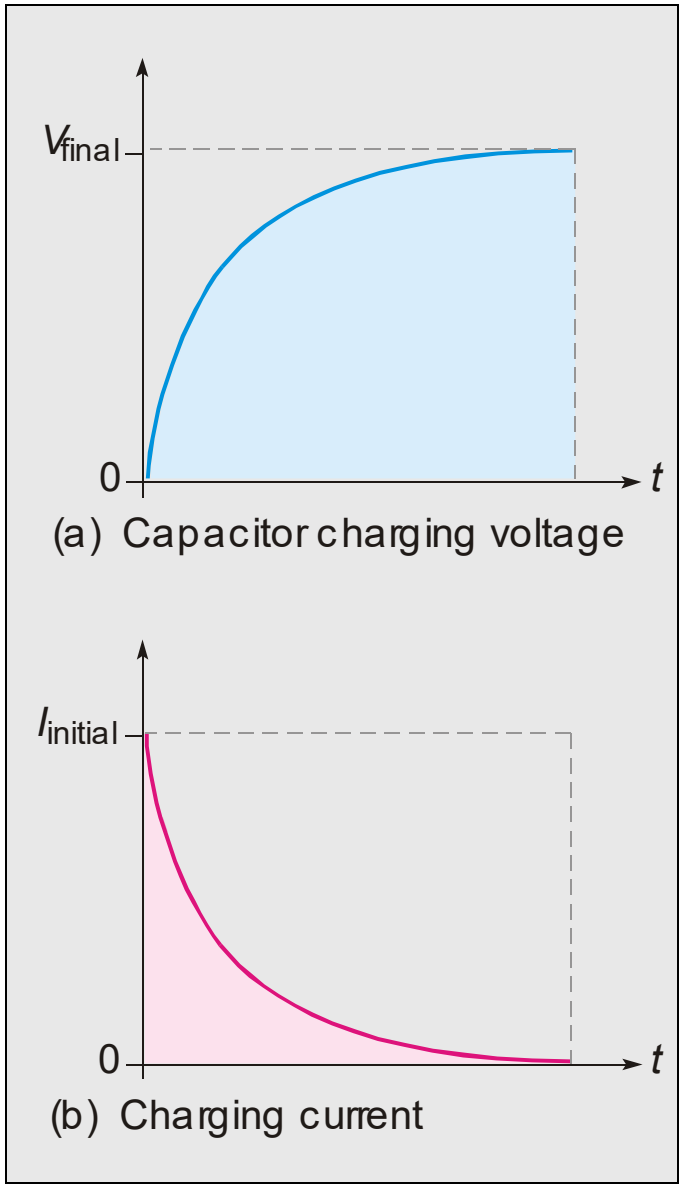


Capacitor Charging:

Capacitor will charge up when it connected to a DC voltage source.

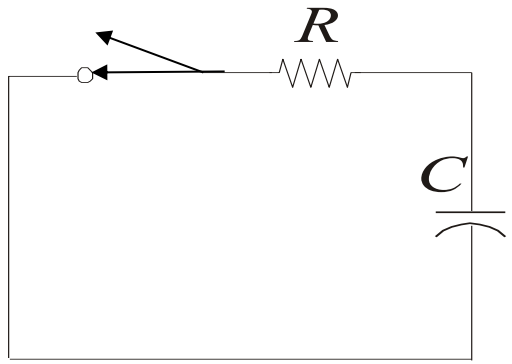
- The current is maximum because the capacitor has zero voltage across it.
- As time passes, the capacitor charges, the current decreases and the voltage across the capacitor increases
- The capacitor charging is exponential.

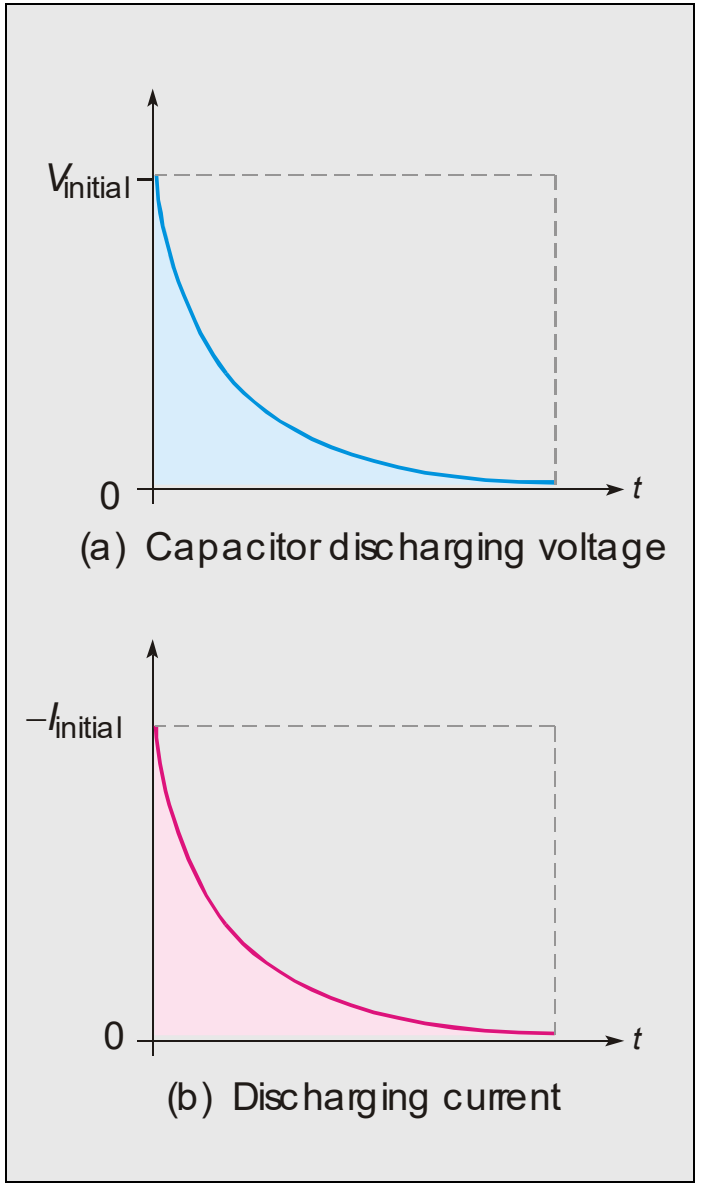




Capacitor Discharging:

- Initially current jumps to maximum, but in opposite direction of charging.
- As time passes, the current and capacitor voltage decreases.
- When a capacitor is discharged through a resistor, the discharge curve is also an exponential. (Note that the current is negative.)





RC Time Constant

Specific values for current and voltage can be read from a universal curve. For an *RC* circuit, the time constant is τ

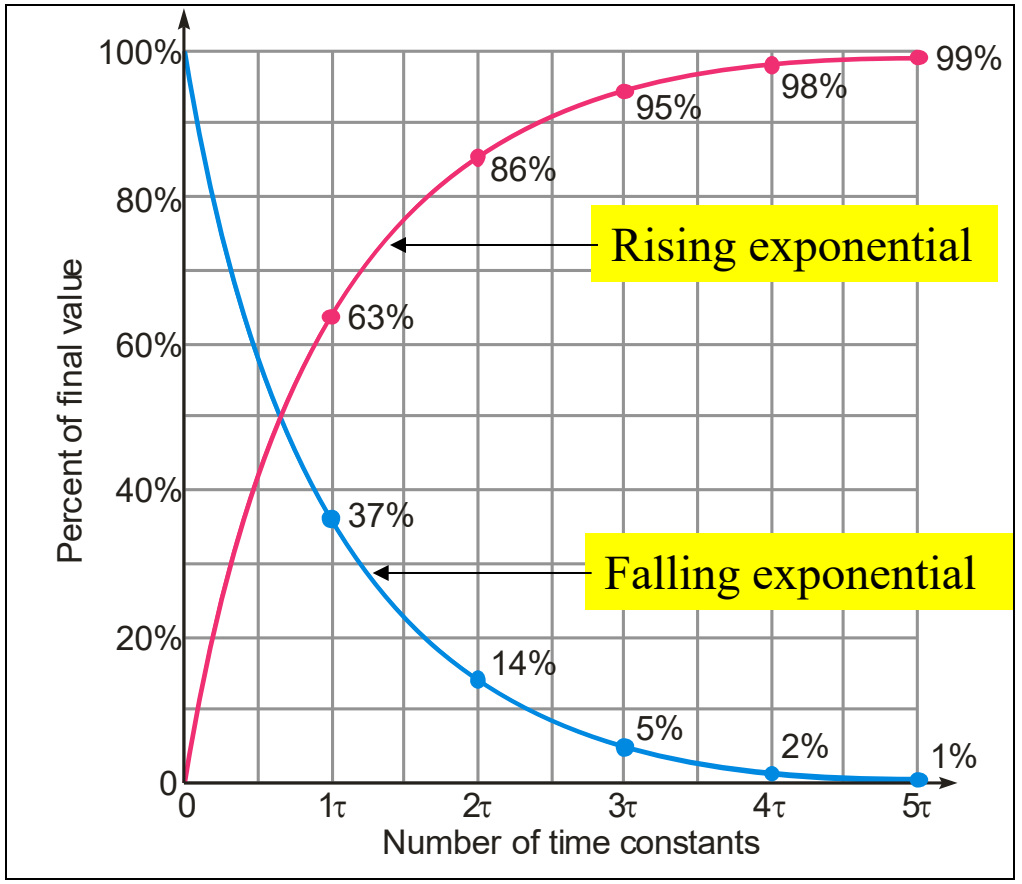
Time constant is the time taken for the voltage to rise or fall by ~63% of its final value.

$$\tau = RC$$

For the rising exponential, 5τ is typically considered to be 100%.

For the falling exponential, 5τ is typically considered to be 0%.

Universal Exponential Curves



Capacitive Reactance

Capacitive Reactance (Xc) is the opposition that a capacitor presents to alternating current (AC) in a circuit. It decreases as the frequency increases.

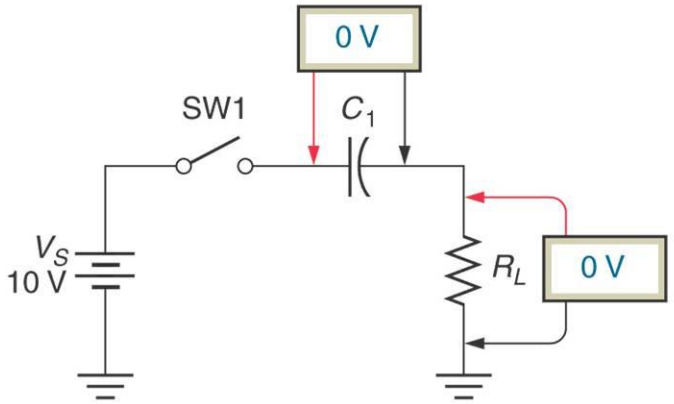
The equation for capacitive reactance is $X_C = \frac{1}{2\pi f C}$

At **low frequencies**, Xc is **high**, meaning the capacitor offers **more opposition** to current flow.
At high frequencies, Xc is **low**, meaning the capacitor allows **more current** to pass through.

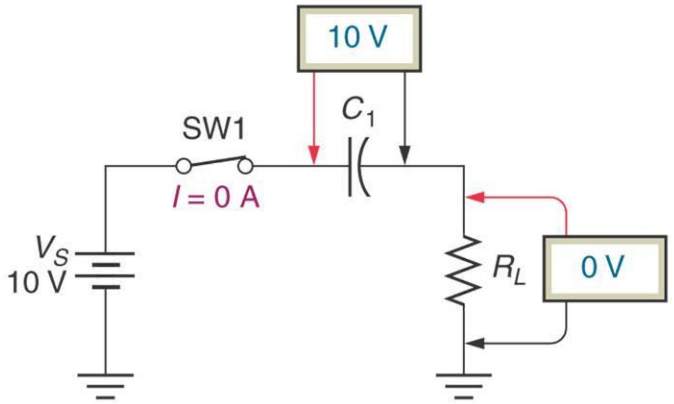
The reactance of a 0.047 μF capacitor when a frequency of 15 kHz is applied is 226 Ω

Capacitor blocks DC

- Capacitor blocks DC **– DC isolation**
- When DC is applied to a capacitor, it **charges** until the voltage across the capacitor equals the applied DC voltage.
- Once fully charged, the capacitor acts like an **open circuit**, blocking further current flow.



(a) Initial circuit conditions

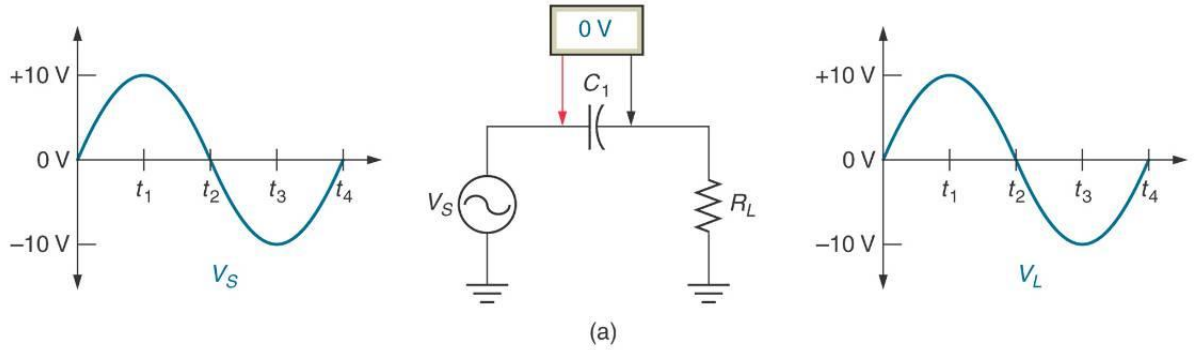


(b) Circuit conditions after the capacitor has charged.

Capacitor passes AC

Capacitor **passes AC**

- When the AC voltage increases, the capacitor charges.
- When the AC voltage decreases and reverses direction, the capacitor discharges and then charges again in the opposite direction.
- This continuous charge-discharge process allows the capacitor to pass the changing AC current.



Capacitive Reactance - Series

When capacitors are in series, the total reactance is the sum of the individual reactances. That is,

$$X_{C(tot)} = X_{C1} + X_{C2} + X_{C3} + \cdots + X_{Cn}$$

Example Assume three 0.033 μF capacitors are in series with a 2.5 kHz ac source. What is the total reactance?

Solution: The reactance of each capacitor is

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(2.5\text{ kHz})(0.033\ \mu\text{F})} = 1.93\ \text{k}\Omega$$

$$X_{C(tot)} = X_{C1} + X_{C2} + X_{C3}$$
$$= 1.93\ \text{k}\Omega + 1.93\ \text{k}\Omega + 1.93\ \text{k}\Omega \quad 5.79\ \text{k}\Omega$$

Capacitive Reactance – Parallel

When capacitors are in parallel, the total reactance is the reciprocal of the sum of the reciprocals of the individual reactances. That is,

$$X_{C(tot)} = \frac{1}{\frac{1}{X_{C1}} + \frac{1}{X_{C2}} + \frac{1}{X_{C3}} + \cdots + \frac{1}{X_{Cn}}}$$

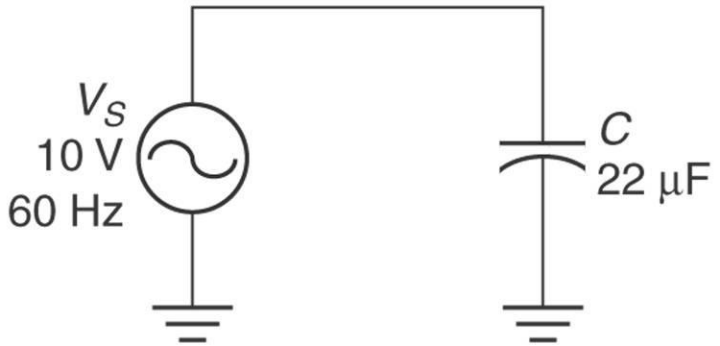
Example If the three 0.033 μF capacitors from the last example are placed in parallel with the 2.5 kHz ac source, what is the total reactance?

Solution: The reactance of each capacitor is 1.93 kΩ

$$X_{C(tot)} = \frac{1}{\frac{1}{X_{C1}} + \frac{1}{X_{C2}} + \frac{1}{X_{C3}}} = \frac{1}{\frac{1}{1.93\ \text{k}\Omega} + \frac{1}{1.93\ \text{k}\Omega} + \frac{1}{1.93\ \text{k}\Omega}} = 643\ \Omega$$

Capacitive Reactance (X_C) and Ohm's Law

- Example: Calculate the total current below

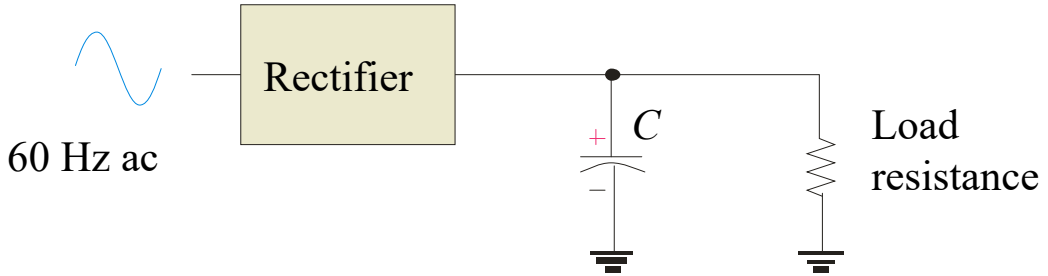


$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(60\text{ Hz})(22\text{ µF})} \cong 121\ \Omega$$

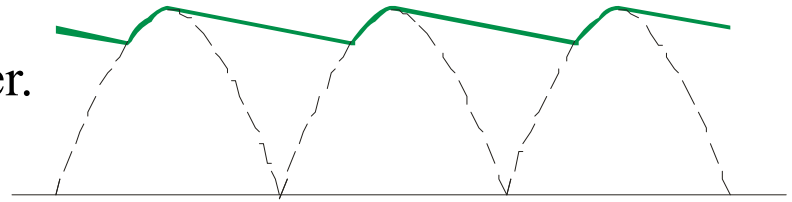
$$I_C = \frac{V_S}{X_C} = \frac{10\text{ V}}{121\ \Omega} = 8.26\text{ mA}$$

Application: Power Supply Filtering

There are many applications for capacitors. One is in filters, such as the power supply filter shown here.



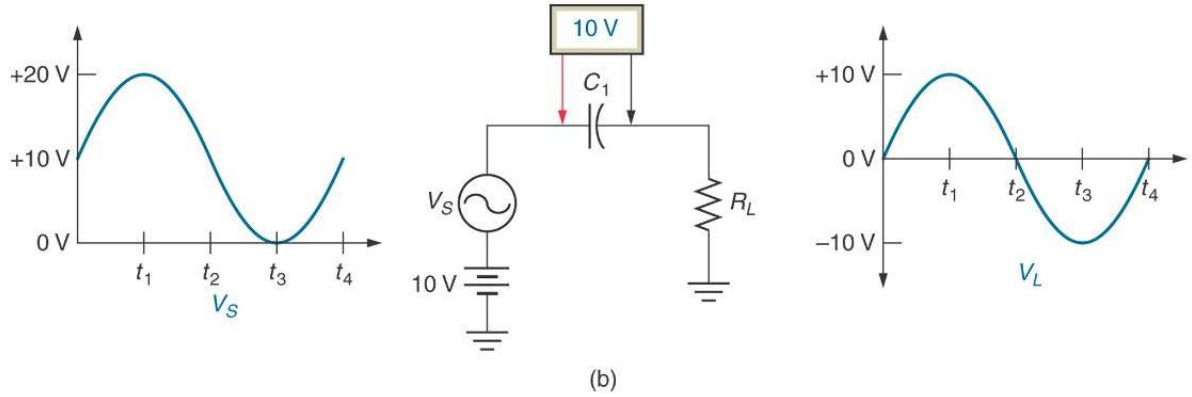
The filter smooths the pulsating dc from the rectifier.



Application: AC Coupling

AC Coupling : AC coupling is used when we need to isolate the AC signal from the DC component, ensuring only the fluctuating signal passes.

- AC signal is coupled – DC offset is blocked



Applications of Capacitors

- **Electrical Storage:**
  Provides energy to smooth out fluctuations in circuits (e.g., power supplies).
- **Powe Supply Filtering:**
  Capacitors remove noise or spikes from power supplies (used in low-pass and high-pass filters).
- **Coupling/Decoupling:**
  - **Coupling:** Passes AC signals while blocking DC in amplifiers.
  - Capacitors block steady DC because they become fully charged and no longer allow current to flow, while they pass AC because the constantly changing voltage keeps them charging and discharging, maintaining current flow.
  - **Decoupling:** Eliminates voltage noise from power lines.
- **Timing Circuits:**
  Works with resistors to create specific delays (RC timing circuits used in clocks, timers).